The unit vector $\hat{a}$ along the direction of $\vec{a}$ is $\frac{\vec{a}}{|\vec{a}|}$ which is equal to $\frac{4\hat{i}-3\hat{j}+10\hat{k}}{5\sqrt{5}}$. The vector whose magnitude is 5 and parallel to $4\hat{i}-3\hat{j}+10\hat{k}$, is $5\left(\frac{4\hat{i}-3\hat{j}+10\hat{k}}{5\sqrt{5}}\right)=\frac{4\hat{i}}{\sqrt{5}}-\frac{3}{\sqrt{5}}\hat{j}+2\sqrt{5}\,\hat{k}$.

So a required point is $\left(\frac{4}{\sqrt{5}},-\frac{3}{\sqrt{5}},2\sqrt{5}\right)$.

Example 8:9

Prove that the points whose position vectors $2\hat{i}+4\hat{j}+3\hat{k}$, $4\hat{i}+\hat{j}+9\hat{k}$ and $10\hat{i}-\hat{j}+6\hat{k}$ form a right angled triangle.

Solution

Let *A, B, C* be the given points and *O* be the point of reference or origin.

Then $\overrightarrow{OA} = 2\hat{i}+4\hat{j}+3\hat{k}$, $\overrightarrow{OB}=4\hat{i}+\hat{j}+9\hat{k}$ and $\overrightarrow{OC}=10\hat{i}-\hat{j}+6\hat{k}$

$$\overrightarrow{AB} = \overrightarrow{OB}-\overrightarrow{OA}=(4\hat{i}+\hat{j}+9\hat{k})-(2\hat{i}+4\hat{j}+3\hat{k})=2\hat{i}-3\hat{j}+6\hat{k}.$$

$$AB = |\overrightarrow{AB}| = \sqrt{2^2+(-3)^2+6^2}=\sqrt{4+9+36}=7$$

$$\overrightarrow{BC} = \overrightarrow{OC}-\overrightarrow{OB}=(10\hat{i}-\hat{j}+6\hat{k})-(4\hat{i}+\hat{j}+9\hat{k})=6\hat{i}-2\hat{j}-3\hat{k}.$$

$$BC = |\overrightarrow{BC}| = \sqrt{6^2+(-2)^2+(-3)^2}=\sqrt{36+4+9}=7$$

$$\overrightarrow{CA} = \overrightarrow{OA}-\overrightarrow{OC}=(2\hat{i}+4\hat{j}+3\hat{k})-(10\hat{i}-\hat{j}+6\hat{k})=-8\hat{i}+5\hat{j}-3\hat{k}.$$

$$CA = |\overrightarrow{CA}| = \sqrt{(-8)^2+5^2+(-3)^2}=\sqrt{64+25+9}=\sqrt{98}$$

$BC^2 = 49$, $CA^2 = 98$, $AB^2 = 49$.

Clearly $CA^2 = BC^2 + AB^2$.

Therefore, the given points form a right angled triangle.

Example 8.10

Show that the vectors $5\hat{i}+6\hat{j}+7\hat{k}$, $7\hat{i}-8\hat{j}+9\hat{k}$, $3\hat{i}+20\hat{j}+5\hat{k}$ are coplanar.

Solution

$$\text{Let } 5\hat{i}+6\hat{j}+7\hat{k} = s(7\hat{i}-8\hat{j}+9\hat{k})+t(3\hat{i}+20\hat{j}+5\hat{k})$$

Equating the components, we have

$$\begin{aligned} 7s+3t &= 5 \\ -8s+20t &= 6 \\ 9s+5t &= 7 \end{aligned}$$

Solving first two equations, we get, $s=t=\frac{1}{2}$, which satisfies the third equation.

Thus one vector is a linear combination of other two vectors.

Hence the given vectors are coplanar.

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